

ARKANA



Odoo V18 Technical Training Day 2

Prasetiya Mulya University

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ODOO V18 • TECHNICAL TRAINING

Day 2

Data Terhubung

Basic Views • Relations • Inheritance • Computed • Onchange

Prasetiya Mulya University
arkana.co.id



Odoo Trainer Technical



Ardian Pramana

Lead Developer @Arkana

Rundown **Training** **Arkana**



- 08.30 - 10.00** : Sesi I
- 10.00 - 10.15** : Coffee Break
- 10.15 - 12.00** : Sesi II
- 12.00 - 13.15** : ISHOMA
- 13.15 - 15.20** : Sesi III
- 15.20 - 15.45** : Break Ashr
- 15.45 - 16.30** : Sesi IV

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CONTINUITY

Previous Checkpoint

Day 1 — Data ada, dan bisa diakses:

- ✓ module academy_management ter-install
- ✓ academy.course & academy.student + fields
- ✓ seed data + menu Academy (Courses, Students)
- ✓ create/edit record dari UI

Dua hal itu yang Day 2 selesaikan.



YANG BELUM

- View masih auto-generated (mentah)
- Model berdiri sendiri tanpa relasi.

Quick Review Day 1

1. Module Structure

Di Odoo custom module, file `__manifest__.py` digunakan untuk apa?

- A. Menyimpan data transaksi user
- B. Mendefinisikan metadata module dan daftar file yang akan di-load
- C. Menjalankan PostgreSQL query secara langsung
- D. Mengatur tampilan form secara otomatis

2. Model dan Database

Ketika kita membuat model:

```
class AcademyCourse(models.Model):  
    _name = "academy.course"
```

Secara konsep, model ini akan merepresentasikan apa?

- A. Menu di UI Odoo
- B. Template QWeb
- C. Business object yang datanya disimpan sebagai tabel di database
- D. File konfigurasi server Odoo

3. Action dan Menu

Kenapa membuat model saja belum cukup agar user bisa membuka datanya dari UI Odoo?

- A. Karena model harus selalu punya computed field
- B. Karena user membutuhkan action dan menu untuk mengakses model dari interface
- C. Karena semua model harus punya wizard
- D. Karena semua data Odoo hanya bisa dibuka dari DBeaver

DAY 2 • GOAL

Target Day 2

Custom views — list / form / search untuk course & student (bukan auto).



Relasi antar-model — academy.batch & academy.enrollment;
Many2one/One2many/Many2many.



Inheritance — model & view inheritance + inherit domain.



Form Ideal — computed (enrollment_count, available_seats) + default + onchange.

Empat bagian: dari tampilan default hingga menjadi form ideal



Part 1 Custom Views

list · form · search



Part 2 Relations

M2o · O2m · M2m



Part 3 Inheritance

model · view · domain



Part 4 Computed · Default · Onchange

form pintar



Posisi journey: Day 1 Data ada → **Day 2 Data terhubung** → Day 3 Data berperilaku → Day 4 Data berbicara → Day 5 Data layak produksi. Lab di tiap Part.

PART 1

Custom Views

Dari auto-generated ke UI yang ideal

PART 1 • BUSINESS PROBLEM

***Form default belum
production ready***

Auto-generated view menampilkan semua field apa adanya. Untuk produk nyata kita butuh layout, urutan, filter, dan search yang didesain sesuai kebutuhan

Tanpa struktur form

Semua field berderet; tak ada group/tab/label rapi.

Tanpa filter & group

List default tak punya search/filter/group_by berguna.

UX seadanya

Tidak siap dipakai user sehari-hari.

Generic view declaration: `view = ir.ui.view`

Kembali ke teori Day 1: view adalah record. Sekarang kita tulis arch XML-nya sendiri.



List & Form views

```
<list> arch
<list>
  <field name="name"/>
  <field name="code"/>
  <field name="level"/>
  <field name="price"/>
</list>
```

```
<form> arch
<form><sheet>
  <group>
    <group>
      <field name="name"/>
      <field name="code"/>
    </group>
    <group>
      <field name="price"/>
    </group>
  </group>
  <notebook>...</notebook>
</sheet></form>
```

Search views: filter & group_by

```
<search> arch
<search>
  <field name="name"/>
  <field name="code"/>
  <filter name="active" string="Active"
    domain="[('active','=',True)]"/>
  <group string="Group By">
    <filter name="grp_level" string="Level"
      context="{ 'group_by': 'level' }"/>
  </group>
</search>
```

<field>

search box untuk mencari record.

<filter> + domain

filter cepat membawa domain preset.

group_by (context)

pengelompokan list lewat context.

Baca source: view XML sale.order

addons/sale/views/sale_order_views.xml · disederhanakan

```
<list>
  <field name="name"/>
  <field name="partner_id"/>
  <field name="amount_total"/>
</list>
<form><sheet>
  <group> ... </group>
  <notebook><page string="Lines">
    <field name="order_line"/>
  </page></notebook>
</sheet></form>
```

list = kolom ringkas

field penting saja untuk overview.

form = sheet/group/notebook

tab 'Lines' menampung One2many.

search = filter + group

filter status + group by customer.

ERP MAPPING pola ditiru view course & batch

Custom views untuk academy.course

```
views/academy_course_views.xml · form
<form><sheet>
  <group>
    <group>
      <field name="name"/>
      <field name="code"/>
      <field name="level"/>
    </group>
    <group>
      <field name="price"/>
      <field name="currency_id"/>
    </group>
  </group>
</sheet></form>
```

list

name · code · level · price.

search

field name/code · filter Active · group_by level.

Student mengikuti pola identik (dikerjakan di Lab A).

ERP MAPPING ≈ Product / Sales Order form

PART 1 • ERP MAPPING

***View Academy =
view Odoo mana
pun.***



Academy course form/list/search — struktur sheet/group/notebook + search filter/group.



Product form (product.template) — katalog dengan tab & filter.



Customer form (res.partner) — person/company dengan group & search.



Sales Order form (sale.order) — header + notebook 'Order Lines'.

Satu pola view, dipakai di seluruh Odoo. Pattern recognition.

PART 1 • LAB CHECKPOINT A

Custom views.

1. Buat views/academy_course_views.xml (list/form/search).
2. Buat views/academy_student_views.xml (pola sama).
3. Set action view_mode = list,form.
4. Daftarkan file di __manifest__.py, lalu -u.
5. Uji filter & group_by di Courses dan Students.

 30 menit • Detail lengkap di Lab Pack — Checkpoint A.

**SELESAI BILA**

- view custom tampil (bukan auto)
- filter & group_by jalan
- student mirip course

Coffee Break

10.00 – 10.15

Waktu Indonesia Barat (WIB)

PART 2

Relations Between Models

Hubungkan data: Batch, Enrollment, Tags

PART 2 • BUSINESS PROBLEM

Master data dan transaksi masih belum terhubung

Course & Student belum saling tahu. Butuh jadwal kelas (Batch), pendaftaran (Enrollment), dan tag pada course.

Tak ada "siapa ikut kelas apa"

Course & Student tak terhubung.

Butuh container jadwal

Batch = kelas terjadwal dari sebuah course.

Butuh pendaftaran (header-line)

Enrollment = student mendaftar ke batch.

Relational fields: Many2one • One2many • Many2many



Many2one

FK ke SATU record.

```
batch.course_id →  
academy.course. Kolom DB  
nyata.
```



One2many

Kebalikan M2o; butuh inverse field.

```
batch.enrollment_ids ←  
enrollment.batch_id. Bukan  
kolom DB.
```



Many2many

Relasi N-N via tabel relasi otomatis.

```
course.tag_ids ↔  
academy.course.tag.
```

Baca source: sale.order ↔ sale.order.line

```
addons/sale/models · disederhanakan
```

```
class SaleOrder(models.Model):  
    _name = "sale.order"  
    partner_id = fields.Many2one(  
        "res.partner")  
    order_line = fields.One2many(  
        "sale.order.line", "order_id")
```

```
class SaleOrderLine(models.Model):  
    _name = "sale.order.line"  
    order_id = fields.Many2one(  
        "sale.order")  
    product_id = fields.Many2one(  
        "product.product")
```

Many2one (line→header)

order_id menunjuk satu sale.order.

One2many + inverse

order_line cermin order_id ('order_id').

header-line pattern

ditiru academy.batch ↔ enrollment.

ERP MAPPING academy.batch ↔ academy.enrollment

Model academy.batch (header)

```
models/academy_batch.py

class AcademyBatch(models.Model):
    _name = "academy.batch"
    name = fields.Char(required=True)
    course_id = fields.Many2one(
        "academy.course",
        ondelete="restrict")
    start_date = fields.Date()
    capacity = fields.Integer()
    state = fields.Selection([...])
    enrollment_ids = fields.One2many(
        "academy.enrollment", "batch_id")
    active = fields.Boolean(default=True)
```

course_id = Many2one

batch milik satu course (onDelete restrict).

enrollment_ids = One2many

inverse 'batch_id' di enrollment.

state, dates, capacity

computed/default menyusul Part 4.

Model academy.enrollment (line)

```
models/academy_enrollment.py  
  
class AcademyEnrollment(models.Model):  
    _name = "academy.enrollment"  
    name = fields.Char(default="New")  
    batch_id = fields.Many2one(  
        "academy.batch",  
        ondelete="cascade")  
    student_id = fields.Many2one(  
        "academy.student",  
        ondelete="restrict")  
    enrollment_date = fields.Date()  
    state = fields.Selection([...])  
    notes = fields.Text()
```

batch_id = Many2one

cascade: hapus batch → hapus enrollment.

student_id = Many2one

restrict: lindungi student yang terdaftar.

embedded di form batch

tampil sebagai inline list (O2m).

Many2many: academy.course.tag

```
models/academy_course_tag.py

class AcademyCourseTag(models.Model):
    _name = "academy.course.tag"
    name = fields.Char(required=True)
    color = fields.Integer()

# tag_ids dipasang ke course via
# MODEL INHERITANCE (Part 3):
tag_ids = fields.Many2many(
    "academy.course.tag")
```

Contoh tags

Python • Backend • Odoo • Integration • Reporting —
satu course bisa banyak tag, satu tag banyak course
(N-N).



Jembatan ke Part 3: Many2many dikenalkan di Part 2; implementasi tag_ids dipasang di Part 3 agar sekaligus menunjukkan model inheritance.

ERP MAPPING ≈ CRM tags / product tags / project tags

PART 2 • ERP MAPPING

Header-line Academy = Sales Order.

Relasi	Academy	Standard Odoo
Many2one	<code>batch.course_id</code>	<code>sale.order.partner_id · line.product_id</code>
One2many	<code>batch.enrollment_ids</code>	<code>sale.order.order_line</code>
Many2many	<code>course.tag_ids</code>	CRM / product / project tags

Pahami satu, kuasai semua: relasi Academy = relasi yang sama di seluruh ERP.

PART 2 • LAB CHECKPOINT B

Hubungkan data.

1. Tambah academy.batch, academy.enrollment, academy.course.tag.
2. Tambah relational fields (M2o/O2m).
3. Tambah ir.model.access.csv untuk 3 model baru.
4. Tambah views + menus + actions Batch & Enrollment.
5. Daftarkan file di manifest, lalu -u academy_management.

 45 menit • Detail lengkap di Lab Pack — Checkpoint B.

**SELESAI BILA**

- menu Batches & Enrollments muncul
- batch menampilkan enrollment lines
- bisa daftarkan student ke batch

ISHOMA

11.50 – 13.05

Waktu Indonesia Barat (WIB)

PART 3

Inheritance

Extend, jangan ubah core

PART 3 • BUSINESS PROBLEM

***Jangan sentuh
source aslinya.***

Butuh menambah field/tampilan ke model yang sudah ada. Mengubah source = mimpi buruk upgrade. Butuh cara aman & terpisah.

**Fork & ubah core**

Rusak saat upgrade; sulit dirawat.

**Extend via inheritance**

Tambah di file terpisah; aman upgrade.

Di proyek nyata: extend res.partner, sale.order, product.template, stock.picking. Academy = sandbox aman.

Model Inheritance (_inherit)

```
academy_course_inherit.py  
  
class AcademyCourse(models.Model):  
    _inherit = "academy.course"  
  
    is_published = fields.Boolean()  
    internal_notes = fields.Text()  
    tag_ids = fields.Many2many(  
        "academy.course.tag")
```



Pakai untuk extend model standar

res.partner
sale.order
product.template
stock.picking

_inherit menambah ke model in-place tanpa membuat tabel baru.

Pola standar: `_inherit` + view XPath

```
module standar meng-extend sale.order

# python
class SaleOrder(models.Model):
    _inherit = "sale.order"
    margin = fields.Monetary()

<!-- xml view inherit -->
<field name="inherit_id"
      ref="sale.view_order_form"/>
<xpath expr="//field[@name=
  'amount_total']" position="after">
  <field name="margin"/>
</xpath>
```

`_inherit` (python)

menambah field ke model standar tanpa fork.

`inherit_id` + `xpath` (xml)

menyisipkan field ke form standar.

pola sejuta module

sale_margin, sale_stock, dsb. pakai ini.

ERP MAPPING kustomisasi sale.order tanpa sentuh source

Model inheritance pada academy.course

```
models/academy_course_inherit.py (file terpisah)
from odoo import models, fields

class AcademyCourse(models.Model):
    _inherit = "academy.course"
    is_published = fields.Boolean(default=False)
    internal_notes = fields.Text()
    tag_ids = fields.Many2many("academy.course.tag")
```

file terpisah

model asli tak disentuh.

is_published · internal_notes

field publishing course.

tag_ids (Many2many)

menyambung Part 2.

Inti: file academy_course.py asli tetap utuh. tag_ids (dari Part 2) akhirnya terpasang di sini. Pola identik dipakai untuk extend res.partner, sale.order, dst.

View Inheritance: inherit_id • xpath • position

```
view inherit
<record id="...form.publishing"
        model="ir.ui.view">
  <field name="inherit_id"

ref="academy_management.view_academy_course_form"/>
  <field name="arch" type="xml">
    <xpath expr="//sheet"
           position="inside">
      <group string="Publishing">...</group>
    </xpath>
  </field>
```

position

inside — di dalam node

after / before — sebelum/sesudah

attributes — ubah properti node

replace — ganti node

PART 3 • ACADEMY CASE — AHA MOMENT

Extend tampilan tanpa sentuh view asli.

```
view inherit: group Publishing + domain
<xpath expr="//sheet" position="inside">
  <group string="Publishing">
    <field name="is_published"/>
    <field name="tag_ids"
      widget="many2many_tags"/>
    <field name="internal_notes"/>
  </group>
</xpath>
<!-- batch: hanya course aktif -->
<xpath expr="//field[@name='course_id']"
  position="attributes">
  <attribute name="domain">
    [('active', '=', True)]</attribute>
</xpath>
```

**Group Publishing**

is_published, tags, internal_notes muncul di form course.

**Inherit domain**

course_id batch hanya tampilkan course aktif.

**Domain ≠ security**

filter di level UI/field, bukan keamanan (Day 3).

PART 3 • LAB CHECKPOINT C

Extend, jangan ubah.

1. Buat models/academy_course_inherit.py (_inherit).
2. Buat inherited form view (xpath group Publishing).
3. Tambah inherit domain (batch.course_id active only).
4. Daftarkan file di manifest, lalu -u academy_management.
5. Uji: arsipkan 1 course → cek dropdown course_id.

 40 menit • Detail lengkap di Lab Pack — Checkpoint C.

**Checkpoint**

- field baru muncul di form course
- view asli tak diubah
- course_id hanya course aktif

Ashar Break

15.10 – 15.45

Waktu Indonesia Barat (WIB)

PART 4

Computed • Default • Onchange

Form yang pintar

PART 4 • BUSINESS PROBLEM

Biarkan data menghitung dirinya.

Hitung jumlah peserta & sisa kursi manual? Isi field yang bisa ditebak sistem? Butuh nilai turunan otomatis, default cerdas, dan bantuan saat input.

Computed field

enrollment_count & available_seats otomatis.

Default value

state=draft, capacity=20, date=today.

Onchange

pilih course → nama batch terisi.

Computed fields & @api.depends

skeleton

```
total = fields.Integer(  
    compute="_compute_total",  
    store=True)  
  
@api.depends("line_ids")  
def _compute_total(self):  
    for rec in self:  
        rec.total = len(rec.line_ids)
```

stored (store=True)

Tersimpan di DB → bisa di-search, group, report.

non-stored (default)

Dihitung saat dibaca; tak bisa di-search/group.

Computed: sale.order.amount_total

```
addons/sale/models/sale_order.py · disederhanakan  
  
amount_total = fields.Monetary(  
    compute="_amount_all", store=True)  
  
@api.depends("order_line.price_subtotal")  
def _amount_all(self):  
    for order in self:  
        order.amount_total = sum(  
            order.order_line.mapped(  
                "price_subtotal")
```

depends pada baris

recompute saat order_line berubah.

store=True

tampil di list & report.

ERP MAPPING $\approx$ batch.enrollment_count / available_seats

Computed: enrollment_count & available_seats

models/academy_batch.py (tambahan)

```
enrollment_count = fields.Integer(  
    compute="_compute_count", store=True)  
available_seats = fields.Integer(  
    compute="_compute_seats", store=True)  
  
@api.depends("enrollment_ids")  
def _compute_count(self):  
    for b in self:  
        b.enrollment_count = len(b.enrollment_ids)  
  
@api.depends("capacity", "enrollment_ids")  
def _compute_seats(self):  
    for b in self:  
        b.available_seats = b.capacity -  
len(b.enrollment_ids)
```

enrollment_count
= len(enrollment_ids).

available_seats
= capacity - count.

store=True
tampil di list, bisa difilter.

Default values & @api.onchange

```
default + onchange  
  
capacity = fields.Integer(default=20)  
date = fields.Date(  
    default=fields.Date.context_today)  
  
@api.onchange("course_id")  
def _onchange_course_id(self):  
    if self.course_id:  
        self.name = self.course_id.name
```

Default values

default=value · lambda · context_today. Mengisi nilai awal record baru.



onchange ≠ write

Onchange bereaksi di form (real-time) tapi hanya in-memory; tersimpan saat user Save.

Defaults + Onchange di Batch & Enrollment

```
models/academy_batch.py & enrollment.py

# batch
state = fields.Selection(..., default="draft")
capacity = fields.Integer(default=20)
@api.onchange("course_id")
def _onchange_course_id(self):
    if self.course_id:
        self.name = self.course_id.name
# enrollment
enrollment_date = fields.Date(
    default=fields.Date.context_today)
```

ERP • Computed

enrollment_count ≈ sale.order.amount_total

ERP • Default

state=draft ≈ Sales Order default state

ERP • Onchange

course_id→name ≈ product_id→price/taxes di SO line

PART 4 • LAB CHECKPOINT D

Form yang pintar.

1. Tambah computed enrollment_count & available_seats.
2. Tambah default (state, capacity, enrollment_date).
3. Tambah onchange course_id → name.
4. Tampilkan computed di list/form batch.
5. Upgrade module & uji di UI.

**Checkpoint**

- count/seats update saat tambah enrollment
- default terisi di record baru
- pilih course mengisi nama batch



40 menit • Detail lengkap di Lab Pack — Checkpoint D.

Jebakan tersering Day 2

**Inverse name salah (One2many)**

enrollment_ids harus tunjuk 'batch_id' yang benar.

**Anggap onchange menulis ke DB**

onchange hanya in-memory sampai Save.

**Domain dikira security**

domain = filter UI, bukan keamanan (Day 3).

**Lupa daftar XML di manifest**

view/menu 'tidak muncul'.

**Lupa dependency @api.depends**

computed bernilai basi / tak update.

**XPath expr salah**

node tak ditemukan → view error. Cek nama field/node.

**Lupa access rights model baru**

batch/enrollment/tag → Access Error.

**Pakai tree (harus list)**

v17+ gunakan list.

Data terhubung – dan mulai enak digunakan



Custom views

list/form/search yang dirancang.



Relations

M2o • O2m • M2m + header-line.



Inheritance

model + view + inherit domain.



Form pintar

computed + default + onchange.



Deliverable Day 2 tercapai: academy_management kini punya custom views, batch & enrollment terhubung, tags, inheritance, dan form pintar.

NEXT · DAY 3

Data berperilaku.

Besok data mulai punya aturan, keamanan, dan interaksi.



Constraints — @api.constrains & SQL constraints — aturan data.



Advanced views — kanban, calendar, pivot, graph, statusbar.



Security — groups, record rules, access control penuh.



Wizards — TransientModel untuk aksi interaktif.

WRAP-UP

Review, QnA & Upcoming Preview

Pertanyaan, blocker lab, dan cuplikan Day 3

Terima Kasih

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